# WELLSCOPE PETROPHYSICAL FORMATION EVALUATION

Well: **WELLSCOPE-SYNTH-01** | Field: NORTH SEA SYNTHETIC | Date: 2026-10-21 | Software Version: 1.0.0

## Executive Summary & Net Pay

| Gross Interval (m) | Reservoir Rock (m) | Net Pay Thickness (m) | Net-to-Gross (NTG) |
|---|---|---|---|
| 150.10 | 69.10 | 34.40 | 22.9% |

## Interpretation Parameters & Assumptions

| Domain | Model | Parameters | Cutoff / Value |
|---|---|---|---|
| Shale Volume (Vsh) | Larionov (Older / Mesozoic) | GR_cl=25.0, GR_sh=115.0 | Vsh <= 0.35 |
| Porosity | Density Porosity | RHOMA=2.65, RHOF=1.0 | PHIE >= 0.1 |
| Saturation | Archie (1942) | Rw=0.05, a=1.0, m=2.0, n=2.0 | Sw <= 0.5 |

## Candidate Pay Intervals

| Zone | Top (m) | Base (m) | Net Pay (m) | Avg Vsh | Avg PHIE | Avg Sw |
|---|---|---|---|---|---|---|
| Zone 1 | 2025.00 | 2040.40 | 15.50 | 0.019 | 0.249 | 0.112 |
| Zone 2 | 2041.10 | 2059.90 | 18.90 | 0.017 | 0.251 | 0.112 |

## Disqualified / Rejected Prospective Intervals

| Zone | Top (m) | Base (m) | Thk (m) | Avg Vsh | Avg PHIE | Avg Sw | Rejection Rationale |
|---|---|---|---|---|---|---|---|
| Excl 1 | 2074.90 | 2075.20 | 0.40 | 0.245 | 0.108 | 0.731 | Water-bearing reservoir sand: Sw avg (0.73) > cutoff (0.50) |
| Excl 2 | 2080.00 | 2109.90 | 30.00 | 0.008 | 0.197 | 0.896 | Water-bearing reservoir sand: Sw avg (0.90) > cutoff (0.50) |

## Quality Control & Borehole Diagnostics

Bit Size: 8.5 in | Washout Threshold: 10.0 in | Washout Samples: 200 | Bad Hole Exclusions: 200

***Disclaimer:*** *WellScope is an educational and portfolio project for conventional wireline log visualization and formation evaluation. Calculated petrophysical properties depend on input quality, curve calibration, lithology, borehole conditions, and user-selected assumptions. Outputs are indicative and must not be used as the sole basis for operational, reserves, completion, or investment decisions.*